Xinrui Wu

xinruiwu.wxr@gmail.com GS Google Scholar Academic GitHub

Education

University of **E**lectronic **S**cience and **T**echnology of **C**hina (**UESTC**) **Sep 2023 - Present**
B.S. in Software Engineering (Embeded System) *Si Chuan, China*
GPA: 3.65/4.0 Average score: 85.01/100

Research Experience

UESTC, CFM (*Center for Future Media*) **Jun 2024 – Present**
Research Intern, Advisor: Prof.Yang Yang ***UESTC**, Sichuan*

- Research focus: advancing **O**pen-**V**ocabulary Object **D**etection (***OVD***) and **S**egmentation (***OVS***).
- Pioneered innovative methods to enhance object detection capabilities, and educe false positives in OVD.
- Collaborated on developing a robust segmentation framework, improving accuracy across various datasets.
- Achieved significant progress in recognizing and segmenting objects with diverse appearances in images.

Megvii Technology Limited **Feb 2025 – Jul 2025**
CV Algorithm Intern, Mentor: Dr.Wei Ge ***MEGVII**, Beijing*

- Proposed unified prompt representation method, transforming multi-modal inputs into visual masks to simplify alignment and improve training efficiency.
- Built large-scale automated data pipeline capable of processing billions of samples to support model training.
- Led development of open-vocabulary detection model achieving performance comparable to closed-source SOTA models.
- Successfully deployed the developed OVD model in business applications, validating its practical utility.

HKUST, VGD Group **Jul 2025 – Aug 2025**
Research Intern, Advisor: Prof.Sai-Kit Yeung ***HKUST**, Hongkong*

- Research focus: lightweight image segmentation and salient individual identification for marine environments.
- Proposed the MaskGuide mask-guided distillation framework, which enhances critical information and suppresses noise through a feature separation and purification process.
- The resulting Tiny-MSAM model achieves 99.4% of the Segment Anything Model's accuracy with only 0.5% of its parameters and an inference speed of 164.7 FPS.
- Accepted by IEEE Robotics and Automation Letters as a co-first-author paper.

Tsinghua, AIR | *Dean: Prof.Ya-qin Zhang* **Jan 2025 – Present**
Research Intern, Advisor: Prof.Yuanchun Li, Prof.Yunxin Liu ***THU**, Beijing*

- Research focus: generation of controllable length **C**hain-**o**f-**T**hought (***CoT***) for large language models.
- Proposed the BudgetThinker framework, centered on an innovative proportional control token mechanism.
- Via a two-stage training pipeline integrating specialized data and reinforcement learning, our model achieved SOTA on benchmarks like MATH with over 95% budget adherence success, and demonstrated generalization in embodied spatial reasoning.
- Accepted by EMNLP 2026 as a co-first-author paper.

MBZUAI | *Dean: Prof.Eric Xing* **Jan 2026 – Present**
Visiting Student, Advisor: Prof.Yutong Xie ***MBZUAI**, Abu Dhabi*

- Research focus: optimization of clinical reasoning.
- Proposed a framework that integrates complex multi-stage clinical information to enhance structured medical reasoning.
- Explored frontier training paradigms for clinical world models.

Publication

BudgetThinker: Empowering Budget-aware LLM Reasoning with Control Tokens

Hao Wen[*], ***Xinrui Wu***[*], *Yi Sun, Feifei Zhang, Liye Chen, Jie Wang, Yifan Su, Yunxin Liu, Ya-Qin Zhang, Yuanchun Li* **2026**

Accepted by ***EMNLP 2026***

MaskGuide: Efficient Distillation for Deployable Lightweight Segmentation in Marine Environments

Xinrui Wu[*], *Ziqiang Zheng*[*], *Yiwei Chen, Zeyu Ma*[+], *Yang Yang, Sai-Kit Yeung* **2026**

Accepted by ***IEEE Robotics and Automation Letters***

Can Image Models Imagine Time? ImageTime: A Novel Benchmark for Probing Visual World Modeling Through Spatiotemporal Consistency

Xinrui Wu[*], *Lichen Huang*[*] **2026**

Accepted by ***WACV 2027***

HiReT: Hierarchical Relational Trajectory Modeling for Longitudinal Clinical Prognosis

Xinrui Wu, *Wanqi Yang, Min Xu, Yutong Xie* **2026**

Submitted to ***ICLR 2027***

Scholarship

UESTC Model Student Scholarship | Top 5% *Sci-Tech Innovation* **Oct 2025**

UESTC Model Student Scholarship | Top 5% *Sci-Tech Innovation* **Oct 2024**

Prize

National Second Prize **Aug 2024**
C4 National College Student Network Technology Challenge

National Third Prize **Jun 2025**
China College Student Computer Design Competition

National Third Prize **May 2025**
National College Student Innovation and Entrepreneurship Competition

National Third Prize **Aug 2024**
C4 National College Student Network Technology Challenge

Second Prize of SiChuan Province **Jun 2024**
China College Student Computer Design Competition

Interests

Clinical AI: Clinical reasoning, longitudinal prognosis, relational trajectory modeling
World Models: Visual world modeling, spatiotemporal consistency, causal reasoning
Efficient Reasoning & Perception: Budget-aware LLM reasoning, lightweight segmentation, distillation
Spatial Intelligence: 3D perception, vision-language spatial reasoning, AR/VR
Generative Models: Controllable image generation, reference-guided editing, generative model evaluation